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QSOX1 confers ferroptosis resistance via redox modification of SLC7A11 in colorectal cancer

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Abstract

Background Ferroptosis, an iron-dependent form of regulated cell death driven by lipid peroxidation, has emerged as a promising therapeutic target for colorectal cancer (CRC). However, the precise mechanisms by which CRC cells evade ferroptosis remain incompletely understood. Cysteine redox modification, typically catalyzed by cysteine oxidases, is a key regulatory factor governing protein structure and function. Quiescin sulphydryl oxidase 1 (QSOX1), a Golgi-localized sulphydryl oxidase known to promote various physiological functions, has an uncharacterized role in ferroptosis. Therefore, this study investigated the effects of QSOX1 on ferroptosis sensitivity in colorectal cancer.

Methods We utilized a comprehensive set of analytical techniques to elucidate the mechanisms of QSOX1 in CRC ferroptosis resistance. We assessed cell proliferation, colony formation, and sensitivity to ferroptosis inducers (Erastin and RSL3) in CRC cells. Levels of reactive oxygen species (ROS), intracellular Fe²⁺, and redox metabolites were measured by biochemical assays and flow cytometry. RNA sequencing (RNA-seq) and untargeted metabolomics were performed using QSOX1-deficient HT-29 cells. The interaction between QSOX1 and SLC7A11 was confirmed via coimmunoprecipitation, immunofluorescence, and nonreducing gel electrophoresis. Cystine uptake and glutamate release assays were used to assess SLC7A11 function. Tumor xenografts were generated to assess the in vivo sensitivity of colon cancer cells to ferroptosis inducers.

Results QSOX1 was significantly upregulated in CRC tissues and promoted CRC cell proliferation in vitro and in vivo. Knockdown of QSOX1 sensitized CRC cells to ferroptosis inducers, whereas QSOX1 overexpression conferred resistance. Mechanistically, QSOX1 enhanced redox homeostasis and GSH metabolism in CRC cells. QSOX1 interacted with and facilitated redox modification of SLC7A11 at cysteine 158 via its thioredoxin motif (C449–C452) in the ERV/ALR domain. This oxidative regulation enhanced SLC7A11 membrane localization, cystine uptake, and glutamate release, thereby boosting intracellular GSH synthesis and suppressing ferroptosis.

Conclusions QSOX1 promotes ferroptosis resistance in CRC cells through redox-dependent post-translational modification of SLC7A11, enhancing its activity and promoting GSH synthesis. The QSOX1–SLC7A11 axis represents a potential therapeutic target to overcome ferroptosis resistance in CRC.



Keywords Colorectal cancer, Ferroptosis, Redox modification, QSOX1, SLC7A11

Introduction

Colorectal cancer (CRC) is a prevalent malignant tumor of the gastrointestinal system, with increasing morbidity and mortality rates in recent years, thereby posing a significant public health challenge [1, 2]. Although surgery-based comprehensive treatment has significantly improved the prognosis of patients with CRC, most patients present with advanced-stage disease at initial diagnosis, precluding the possibility of curative resection. While chemotherapy, targeted therapy, and immunotherapy have improved clinical outcomes for certain patient subtypes with advanced CRC, drug resistance and disease recurrence remain significant challenges in CRC treatment. These limitations underscore the urgent need to elucidate the molecular pathological mechanisms underlying CRC progression and drug resistance to discover novel therapeutic targets.

CRC progression is often accompanied with disturbances in glucose metabolism, lipid metabolism, amino acid metabolism, and oxidative phosphorylation, leading to an imbalance in cellular redox homeostasis [3]. Ferroptosis, a form of metabolic dysregulation-driven cell death, has garnered significant attention in recent years. Furthermore, ferroptosis differs from apoptosis, necrosis, and other cell death modalities, primarily characterized by iron-dependent accumulation of lipid peroxides (LPO) and depletion of polyunsaturated fatty acids in cellular membranes [4]. A key mechanism of ferroptosis is the induction of cell death via lipid peroxidation of unsaturated fatty acids in cellular membranes, mediated by divalent iron or lipoxygenases (LOXs) [5]. An important component of ferroptosis regulation is the System Xc⁻ antiporter, a heterodimeric complex comprising solute carrier family 7 member 11 (SLC7A11, also known as xCT) and solute carrier family 3 member 2 (SLC3A2, also known as CD98 or 4F2hc) [6]. This antiporter facilitates the uptake of cystine into the cell in exchange for intracellular glutamate. Intracellularly, cystine is reduced to cysteine, the rate-limiting precursor for glutathione (GSH) synthesis. GSH is essential for the function of glutathione peroxidase 4 (GPX4), a key enzyme that detoxifies lipid peroxides, thereby inhibiting ferroptosis [7]. SLC7A11 is highly expressed in various cancer types, including CRC, and it is associated with increased antioxidant capacity, ferroptosis resistance, and poor prognosis [8]. Dysregulation of ferroptosis has been implicated in several cancers, including CRC [3, 9]. Tumor cells often develop resistance to ferroptosis, enabling them to evade oxidative stress and thrive in unfavorable microenvironments. However, the mechanisms by which CRC cells resist ferroptosis warrant further investigation.

Disulfide bonds are crucial for protein structural stability and are prevalent in secreted and membrane proteins, playing a key role in the regulation of protein function. The formation of disulfide bonds depends on the catalytic activity of sulfhydryl oxidase, which is widely distributed in the endoplasmic reticulum, other cellular organelles, and extracellular spaces [10]. Quiescin sulfhydryl oxidase 1 (QSOX1) is a unique flavin-dependent disulfide oxidase that directly catalyzes the oxidation of sulfhydryl groups in substrate proteins, forming disulfide bonds and promoting their proper folding [11]. Unlike traditional protein disulfide isomerases, which are exclusively localized to the endoplasmic reticulum, QSOX1 is present not only in the endoplasmic reticulum but also in the Golgi apparatus, secretory vesicles, and the extracellular space [12]. Furthermore, QSOX1

is highly expressed in various tumors and promotes tumor cell proliferation, invasion, and metastasis, potentially through tumor microenvironment regulation and extracellular matrix remodeling [13, 14]. However, the expression and mechanisms of QSOX1 in CRC remain poorly understood. Thiol oxidation of proteins, particularly disulfide bond formation, directly affects protein conformation, stability, and function. Given that SLC7A11 is a transmembrane protein with an extracellular loop region rich in cysteine residues, it may be a target for disulfide bond redox modification, which could regulate its protein stability and transport activity. However, whether QSOX1 regulates ferroptosis-related proteins, particularly SLC7A11, through oxidative modification to affect ferroptosis resistance of tumor cells remains unclear.

In the present study, we found that QSOX1 promotes the functional activity of SLC7A11 through oxidative modification, thereby enhancing the ferroptosis resistance of CRC cells. This mechanism not only expands our understanding of the role of oxidative modification in the ferroptosis regulatory network but also provides new molecular targets for overcoming tumor resistance to ferroptosis. Targeting the QSOX1–SLC7A11 axis may represent a promising new strategy for future CRC therapy.

Materials and methods

Antibodies and reagents

Antibodies against QSOX1 (No. 12713-1-AP), SLC7A11 (No. 26864-1-AP), SLC3A2 (No. 15193-1-AP), GAPDH (1E6D9) (No. 60004-1-Ig), Flag-tag (8H6A10) (No. 66008-4-Ig), Myc-tag (No. 16286-1-AP), PARP1 (No. 13371-1-AP), Cleaved caspase 3 (No. 25128-1-AP), MLKL (No. 21066-1-AP), GPX4 (3F5G5) (No. 67763-1-Ig), LC3 (No. 14600-1-AP), and β -actin (No. 20536-1-AP) were obtained from Proteintech. Antibody against SLC7A11 (No. MA5-44962) was obtained from Thermo Fisher Scientific. Antibody against the HA-tag (No. M132) was obtained from MEDICAL & BIOLOGICAL LABORATORIES CO., LTD. Antibody against Phospho-MLKL (Ser358) (D6H3V) (#91689 T), Caspase-3 (#9662), and alpha-Tubulin (11H10) (#2125) was obtained from Cell Signaling Technology. HRP-labeled anti-mouse IgG (No. 5220-0338) and HRP-labeled anti-rabbit IgG (No. 5220-0335) were obtained from SeraCare. Alexa Fluor Plus 488-conjugated Goat anti-Rabbit IgG (No. A-11008) and Alexa Fluor Plus 647-conjugated Goat anti-Mouse IgG (No. A-21235) were obtained from Thermo Fisher Scientific. Erastin (No. HY-15763), RSL3 (No. HY-100218A), SBI-183 (No. HY-165245), Z-VAD-FMK (HY-16658B), Necrostatin-1 (HY-15760), and Ferrostatin-1 (HY-100579) were purchased from MedChemExpress. Chloroquine (C6628) was purchased from Sigma-Aldrich.

Cell lines and cell culture

Human CRC cell lines HT-29 and HCT116 were maintained in McCoy's 5A medium (Macgene). The SW480 and HEK293T cells were maintained in Dulbecco's modified eagle medium (DMEM; Macgene). All cells were purchased from the American Type Culture Collection. All cells were supplemented with 10% fetal bovine serum (VivaCell), 100 U/mL of penicillin, and 100 μ g/mL of streptomycin. All cells were cultured at 37 °C in a humid environment (5% CO₂–95% air). All cells were harvested via brief incubation in 0.25% (w/v) ethylenediaminetetraacetic acid (EDTA)-phosphate-buffered saline (PBS).

Human tissue specimens

CRC cases in the First Affiliated Hospital of Shandong First Medical University were selected. The research was approved the Medical Ethics Committee of the First Affiliated Hospital of Shandong First Medical University (YXLL-KY-2023-002) and all patients provided informed consent. Details of patient information are listed in Supplementary Table 1.

Transient transfection of plasmids and siRNAs

The cells underwent transient transfection with pcDNA3.1, pcDNA3.1-QSOX1, and pcDNA3.1-SLC7A11 using jetPRIME (Polyplus) for 48 h in accordance with the manufacturer's instructions. Human QSOX1 siRNA and human SLC7A11 siRNA (GenePharma) were transfected into the cells using INTERFERin (Polyplus) for 48 h for the siRNA experiment. Scrambled siRNA functioned as a control. After transfection, the cells underwent therapy. Three independent experiments were conducted. The siRNA sequences are listed in Supplementary Table 2.

Western blotting assay

Protein expression in cells was analyzed via western blotting. Cells were first rinsed twice with ice-cold PBS and subsequently lysed with m-PER lysis buffer (Thermo Fisher) containing a freshly prepared protease inhibitor cocktail (Solarbio). Protein concentrations were determined using a BCA protein assay kit (Beyotime Biotechnology). Lysate samples containing equal amounts of protein were denatured by the addition of 5× sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE) sample buffer containing a reducing agent, separated by SDS-PAGE, and transferred to nitrocellulose (NC) membranes (Cytiva).

For nonreducing PAGE, samples were denatured using 5× SDS-PAGE sample buffer without a reducing agent. The membranes were blocked with 5% nonfat dried milk in TBS (20 mM Tris-HCl, pH 7.6, 150 mM NaCl) at room temperature for 1 h and incubated with primary antibody at 4 °C overnight. Thereafter, the membranes were incubated with HRP-conjugated secondary antibody at room temperature for 1 h, followed by three washes with TBS-T. Proteins were visualized using an enhanced chemiluminescence detection system.

Hematoxylin and eosin (H&E) staining and immunohistochemical analysis

Formalin-fixed, paraffin-embedded tissue sections were deparaffinized and rehydrated using a graded ethanol series. H&E staining was performed using standard procedures, with hematoxylin for nuclear staining and eosin for cytoplasmic staining. Afterward, the sections were dehydrated, cleared, and mounted with neutral resin.

For immunohistochemistry, sections underwent antigen retrieval in Tris-EDTA buffer, followed by blocking of endogenous peroxidase activity with hydrogen peroxide. After washing, sections were incubated with the appropriate primary antibody at 4 °C overnight. Thereafter, sections were incubated with HRP-conjugated secondary antibody. Following DAB chromogen development and hematoxylin counterstaining, images were captured using a microscope (Nikon). Nuclei stained blue, cytoplasm stained red, and areas of positive protein expression stained brown.

Cell viability assay

Cells were seeded in 96-well plates at a density of 1×10^4 cells/well. Following treatment, 10 μ L of MTT reagent (5 mg/mL) was added to each well, and the cells were incubated at 37 °C for 2 h. Subsequently, the MTT-containing medium was carefully aspirated, 100 μ L of DMSO was added to each well to dissolve the formazan crystals, and the absorbance was measured at 570 nm using a microplate reader.

Colony formation assay

Cells were seeded in six-well plates at a density of 500 cells per well and cultured at 37 °C for 2 weeks. Cells were fixed with 4% paraformaldehyde and stained with Giemsa dye.

Immunofluorescence staining

Cells were seeded on coverslips, treated, and fixed with 4% paraformaldehyde in PBS at room temperature for 15 min. Following fixation, cells were permeabilized with PBS containing 0.1% Triton X-100 and then blocked with PBS containing 3% bovine serum albumin for 1 h. After washing, cells were incubated with the appropriate primary antibody at 4 °C overnight. Following washing, cells were incubated with a fluorophore-conjugated secondary antibody, and nuclei were counterstained with DAPI (Thermo Fisher). Fluorescence images were acquired using a confocal microscope (Carl Zeiss).

Co-immunoprecipitation

Treated cells were lysed with IP buffer (50 mM Tris–HCl (pH 7.4), 150 mM NaCl, 1% NP-40, 0.2 mM EDTA, 10 mM N-ethylmaleimide, and protease inhibitor cocktail). Protein concentrations of the lysates were determined using a BCA protein assay kit. Subsequently, the lysates were incubated with the target primary antibody at 4 °C overnight on a rotating platform.

The following day, protein G-conjugated magnetic beads (Thermo Fisher) were added and incubated for 1 h. Thereafter, the beads were washed with wash buffer (50 mM Tris–HCl (pH 7.4), 150 mM NaCl, 1% NP-40, 0.2 mM EDTA) for 30 min. Afterwards, the magnetic beads were washed five times for 5 min each. Bound proteins were eluted with glycine (0.1 M, pH 2.0), immediately neutralized with 1 M Tris (pH 8.0), and denatured by adding SDS-PAGE sample buffer for immunoblotting analysis.

Drug combination analysis

To assess the combined effects of Erastin/RSL3 and SBI-183, cells were treated with varying concentrations of each drug—alone or in combination—for 48 h, with three replicates per condition. Drug synergism was evaluated using CompuSyn software (Biosoft, Ferguson, MO, USA). Combination index (CI) values were calculated according to the software's instructions. $CI < 1$, $CI = 1$, and $CI > 1$ were defined as indicative of synergistic, additive, and antagonistic effects, respectively.

Reactive oxygen species (ROS) measurement

Treated cells were incubated with 10 μ M 2',7'-dichlorodihydrofluorescein diacetate (DCFH-DA, Beyotime) to monitor intracellular ROS accumulation. Cells loaded with DCFH-DA were incubated at 37 °C for 30 min, with gentle agitation every 5 min to ensure adequate probe penetration. Subsequently, cells were washed, resuspended in

PBS, and immediately analyzed via flow cytometry (Beckman Coulter) using the fluorescein isothiocyanate (FITC) channel to detect ROS levels.

Hydrogen peroxide (H₂O₂) measurement

Hydrogen peroxide (H₂O₂) levels were quantified using the Hydrogen Peroxide Assay Kit (Beyotime Biotechnology). For cultured cells, the cells were first harvested into centrifuge tubes, and the supernatant was discarded. Lysis buffer was added at a ratio of 100–200 µL per 1×10^6 cells, followed by thorough homogenization to ensure complete cell lysis. The lysates were centrifuged at approximately $12,000 \times g$ at 4 °C for 5 min, and the resulting supernatant was collected. Simultaneously, an H₂O₂ standard curve was prepared. The H₂O₂ detection reagent was thawed on ice or in an ice bath. Subsequently, 50 µL of sample or standard was added to each detection well or tube, followed by the addition of 100 µL of the detection reagent. The mixture was incubated at room temperature for 30 min. Absorbance was immediately measured at 560 nm, and the H₂O₂ concentration in the samples was calculated on the basis of the standard curve.

Lipid peroxidation MDA assay

Malondialdehyde (MDA) content was quantified using the Lipid Peroxidation MDA Assay Kit (Beyotime Biotechnology). For cell lysis, cultured cells were lysed using mPER Mammalian Cell Lysis Buffer (Thermo Fisher) at a ratio of 100 µL per 1×10^6 cells. Following lysis, the samples were centrifuged at $12,000 \times g$ for 10 min, and the resulting supernatant was collected. Concurrently, the thiobarbituric acid (TBA) working reagent and MDA standard curve were prepared. The lysate was then mixed with the reaction reagent, incubated in a boiling water bath for 15 min, cooled to room temperature in a water bath, and subsequently centrifuged at $1000 \times g$ for 10 min at room temperature. Approximately 200 µL of the supernatant were then transferred to a 96-well plate, and the absorbance was measured at 532 nm using a microplate reader. The final MDA content was normalized to the total protein concentration.

Intracellular Fe²⁺ assay

Intracellular Fe²⁺ levels were measured using a Colorimetric Fe²⁺ Assay Kit (Beyotime Biotechnology) according to the manufacturer's instructions with minor modifications. Briefly, cells were collected by centrifugation at $500 \times g$ for 5 min. After removal of the supernatant, cell pellets were lysed in lysis buffer at a ratio of 100 µL per 1×10^6 cells, homogenized, and incubated on ice for 10 min. The lysates were centrifuged at $12,000 \times g$ for 5 min at 4 °C, and the supernatants were collected. For acidification, 5 µL of 1 M HCl was added to each 100 µL of sample, followed by incubation at 60 °C for 30 min. The samples were then centrifuged at $12,000 \times g$ for 10 min at 4 °C. After preparation of the iron standard curve, samples were mixed with Decontamination Solution and incubated at 37 °C for 30 min. Subsequently, 10 µL of Iron Probe was added to each well, and the reaction mixture was incubated at 37 °C for 30 min in the dark. Absorbance was measured at 593 nm using a microplate reader, and intracellular Fe²⁺ levels were calculated on the basis of the standard curve.

Fluorescent probe detection of cellular Fe²⁺

The Fe²⁺ probe, FerroOrange, was purchased from DOJINDO LABORATORIES. FerroOrange was prepared by adding 35 μ L of DMSO to 24 μ g of FerroOrange and mixing thoroughly by pipetting. This stock solution was subsequently diluted in HBSS solution to generate a 1 μ mol/L working solution. Cells were seeded onto fluorescence microscopy dishes and cultured overnight. Prior to detection, the supernatant was aspirated, and the cells were washed three times with HBSS. The medium was then replaced with drug-containing medium for the corresponding treatment duration. Following treatment, the medium was removed, and the cells were washed three times with HBSS. Finally, the cells were incubated with the 1 μ mol/L FerroOrange working solution in the incubator for 30 min before being visualized using fluorescence microscopy.

Measurement of cystine uptake

Cystine uptake was measured using the Cystine Uptake Assay Kit (Dojindo) following the manufacturer's protocol. Cells in a black 96-well plate were incubated with cystine analog solution at 37 °C for 30 min, washed with cold PBS, and then lysed. Fluorescence (Ex/Em = 490/535 nm) was measured using a Tecan plate reader after adding the working solution. Blank values were subtracted for quantification, with optional protein normalization.

GSH assay

The GSH concentration of the cell lysate was measured using an assay kit from Nanjing Jiancheng (#A006-2). Cells were lysed in PBS or directly in the provided precipitation reagent, and the lysate was centrifuged to obtain a clear supernatant. Afterward, the supernatant was mixed with 5,5'-dithiobis-(2-nitrobenzoic acid) and buffer, which reacts with GSH to produce a yellow-colored compound. The absorbance of this product was measured at 405 nm using a microplate reader and normalized to the total protein concentration.

Glutamate assay

Extracellular glutamate levels in cell culture supernatants were quantified using a commercial glutamate assay kit (Nanjing Jiancheng, #A074-1-1). The assay measures glutamate via enzymatic conversion to α -ketoglutarate and NADH, detected spectrophotometrically at 340 nm. Samples were diluted at a 1:3 ratio with a reagent, centrifuged, and reacted with a working solution before being incubated at 37 °C for 40 min. Absorbance differences were used to calculate glutamate concentration, normalized to total protein content from cell lysates to ensure consistency.

Tumor xenograft experiment

Female BALB/c nude mice (6 weeks of age) were obtained from Beijing HFK Bioscience Co., Ltd. (Beijing, China). After a week of acclimatization, control and QSOX1 knock-down HT-29 cells (5×10^6 cells) were suspended in 100 μ L saline and injected subcutaneously into the right axilla of the mice. When tumors reached 50–100 mm³, mice were administered intraperitoneal injection placebo or Erastin (25 mg/kg) daily.

After 14 days of treatment, all mice were sacrificed, and tumors were removed and weighed. All experimental protocols were approved by the Institutional Animal Care

and Use Committee of the First Affiliated Hospital of Shandong First Medical University (2024010901).

RNA-seq analysis

Total RNA was extracted from HT-29 cells with stable QSOX1 depletion and their corresponding control cells. The cDNA library was obtained and confirmed using the Agilent 2100 bioanalyzer (Agilent Technologies, USA). Sequencing was performed using the Illumina NovaSeq X Plus (Illumina, USA). The library creation and sequencing were conducted at Novogene Bioinformatics Institute Co., Ltd. (China). Cuffdiff was used to assess differentially expressed genes (DEGs). Log2 fold change > 1 and P -value < 0.05 were used to identify DEGs. Heatmap was plotted by <https://www.bioinformatics.com.cn>, an online platform for data analysis and visualization.

Untargeted metabolomics relative-quantitative analysis

Control and QSOX1 knockdown HT-29 cells were thawed on ice and then combined with 1 mL of ice-cold methanol/acetonitrile/water (2:2:1, v/v/v). The homogenate was sonicated on ice for 30 min (two cycles), followed by centrifugation at 4 °C and 14,000 g for 20 min. The supernatant was collected and dried using a vacuum centrifuge.

For liquid chromatography–mass spectrometry (LC–MS) analysis, samples were reconstituted in 100 µL of acetonitrile/water (1:1, v/v). Analyses were performed using a quadrupole time-of-flight mass spectrometer (AB Sciex TripleTOF 6600) coupled to an ultra-high-performance liquid chromatography system (Agilent 1290 Infinity LC) at Shanghai Applied Protein Technology Co., Ltd.

Statistical analysis

Data are presented as mean ± SD. Unless otherwise stated, all data shown are representative of at least three independent experiments. Statistical analyses were performed using GraphPad Prism 8. Data were analyzed by either a two-tailed Student's t -test (P < 0.05 considered statistically significant) or a one-way analysis of variance (ANOVA) with Tukey's or Dunnett's multiple comparisons test (P < 0.033 considered statistically significant), as indicated.

Results

QSOX1 is essential for the proliferation of CRC cells

The expression and function of QSOX1 are tumor-specific [11]. To investigate the expression of QSOX1 in CRC, we collected 16 pairs of tumor tissues and adjacent noncancerous tissues from patients with CRC. Western blotting analysis showed that the protein expression of QSOX1 was higher in tumor tissues than in noncancerous matched tissues (Fig. 1A).

Immunohistochemistry was conducted to detect the expression and tissue distribution of QSOX1 in CRC and adjacent noncancerous tissues. The staining intensity of QSOX1 in tumor tissues was significantly higher than that in adjacent noncancerous tissues (Fig. 1B). Moreover, QSOX1 was specifically overexpressed in tumor tissues, with lower expression in the mesenchyme (Fig. 1B). QSOX1 can be secreted into the extracellular space, where it regulates the extracellular matrix and modulates anti-tumor immune response [15]. Enzyme-linked immunosorbent assay (ELISA) results showed that serum

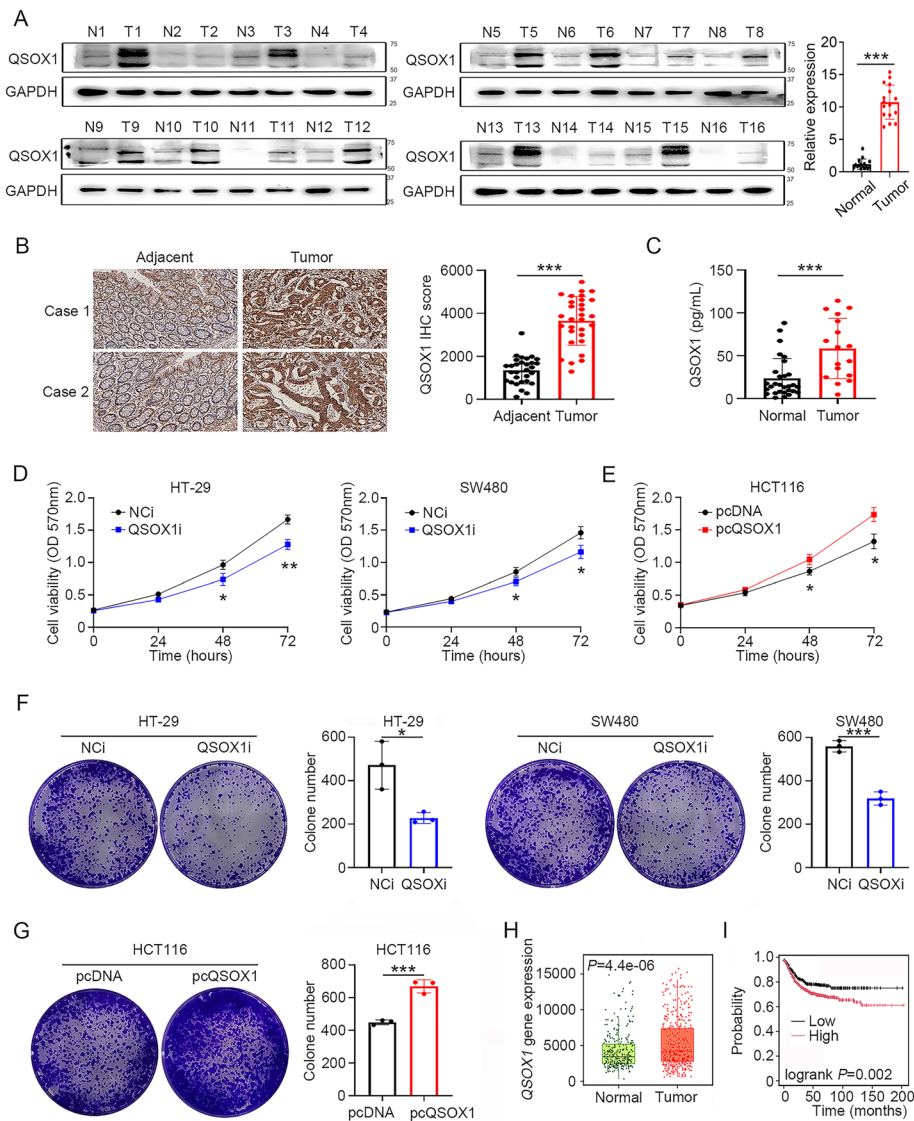


Fig. 1 QSOX1 is essential for the proliferation of colorectal cancer (CRC) cells. **A** Western blotting analysis of QSOX1 in CRC tissue samples ($n=16$) compared with adjacent noncancerous tissues ($n=16$). **B** QSOX1 expression in CRC tissues and adjacent noncancerous tissues was detected via immunohistochemistry. **C** Enzyme-linked immunosorbent assay (ELISA) analysis of QSOX1 in the serum of patients with CRC ($n=17$) and healthy controls ($n=29$). **D** and **E** MTT assay of proliferative activity in HT-29 and SW480 cells treated with siRNA targeting QSOX1 **D** and HCT116 cells transfected with QSOX1 expression plasmids **E** ($n=3$). **F** and **G** Clonogenic assay of proliferative activity in HT-29 and SW480 cells transfected with QSOX1 siRNA **F** and HCT116 cells transfected with QSOX1 expression plasmids **G** ($n=3$). **H** Boxplots assessing QSOX1 expression between colon cancer and healthy tissues based on the TNMplot Database. **I** QSOX1 is correlated with poor prognosis in CRC. Data in **B–G** are presented as mean $\pm$ SD and P -values obtained via two-tailed unpaired Student's t -test are indicated. *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$

QSOX1 levels were higher in patients with CRC than in healthy controls (Fig. 1C). These findings suggest that QSOX1 may function as an oncogene, promoting CRC progression.

To validate this hypothesis, we first examined the basal expression levels of QSOX1 across the colorectal cancer cell lines HT-29, SW480, and HCT116. The results revealed significantly higher QSOX1 expression in HT-29 and SW480 cells relative to HCT116 cells (Supplementary Fig. 1A and 1B). Then we used small interfering RNA (siRNA) to knock down QSOX1 expression in HT-29 and SW480 cells, which express high levels of QSOX1 (Supplementary Fig. 1C–1F). As shown in Fig. 1D, cell proliferation assays

demonstrated that QSOX1 inhibition significantly suppressed the proliferative viability of HT-29 and SW480 cells. Similarly, QSOX1 overexpression in HCT116 cells, which express low levels of QSOX1, significantly increased cell proliferative capacity (Fig. 1E, Supplementary Fig. 1G and 1H). Furthermore, QSOX1 knockdown significantly inhibited colony formation in HT-29 and SW480 cells, whereas QSOX1 overexpression promoted colony formation in HCT116 cells (Fig. 1F and G).

We further validated our results using public databases. The TNMplot database [16] showed that *QSOX1* gene expression was significantly higher in colon cancer tissues than in healthy colon tissues (Fig. 1H). Kaplan–Meier analysis [17] indicated that patients with colon cancer with high QSOX1 expression had a poor prognosis (Fig. 1I). These data suggest that QSOX1 promotes CRC progression.

QSOX1 promotes CRC progression by regulating cellular redox homeostasis

To elucidate the molecular mechanism by which QSOX1 facilitates CRC progression, we conducted RNA-seq analysis to characterize the transcriptomic alterations in QSOX1-deficient HT-29 cells. In total, 294 DEGs were identified in comparison with the control group (FDR value < 0.05 and $|\log_2(\text{Fold change})| > 1$), with 217 genes upregulated and 77 genes downregulated (Fig. 2A and B).

Gene ontology (GO) enrichment analysis of the biological process (BP) showed that the upregulated genes were associated with BPs related to the unfolded protein response, oxygen sensing, and extracellular structure organization, whereas the downregulated genes were involved in ammonium ion metabolism (Fig. 2C). GO enrichment analysis of the molecular functions (MF) of DEGs revealed that upregulated genes were enriched for protein folding, whereas downregulated genes were enriched for kinase activity and receptor activity (Fig. 2D). Kyoto Encyclopedia for Genes and Genomes (KEGG) pathway analysis indicated that upregulated genes were involved in HIF-1 signaling, the MAPK signaling pathway, and other pathways, whereas downregulated genes were enriched in metabolic and DNA repair (Fig. 2E). As shown in Fig. 2F, protein–protein interaction (PPI) analysis revealed that the interaction network of DEGs was enriched in protein folding, oxygen sensing, and metabolic pathways.

These data suggest that QSOX1 may play a role in metabolic homeostasis and protein synthesis quality control in CRC cells. Given that QSOX1 is an important sulfhydryl oxidase in disulfide bond formation, it can regulate hydrogen peroxide (H_2O_2) production in cells [18]. We analyzed the expression of redox-related genes in the RNA-seq data. In QSOX1-knockdown HT-29 cells compared with control cells, the expression of genes involved in redox homeostasis, such as *ERO1A*, *NFE2L2*, and *PRDX1*, was upregulated, whereas the expression of genes associated with cystine transport and iron homeostasis, including *SLC7A11*, *SLC3A2*, and *FTH1*, was downregulated (Fig. 2G). Therefore, we hypothesized that QSOX1 is associated with cellular redox homeostasis. QSOX1 promotes ROS production in hepatocellular carcinoma cells [19]; however, its role in CRC remains unclear. We used a ROS probe to measure ROS levels in HT-29 and SW480 cells with QSOX1 knockdown using siRNAs. The results showed that ROS levels were significantly elevated in QSOX1 knockdown cells (Fig. 2H). In contrast, ROS levels in QSOX1-overexpressing HCT116 cells were significantly lower than the levels in the control group (Fig. 2I). Because QSOX1 generates H_2O_2 during the catalysis of disulfide bond formation in its substrate proteins [11], we further evaluated whether H_2O_2 levels

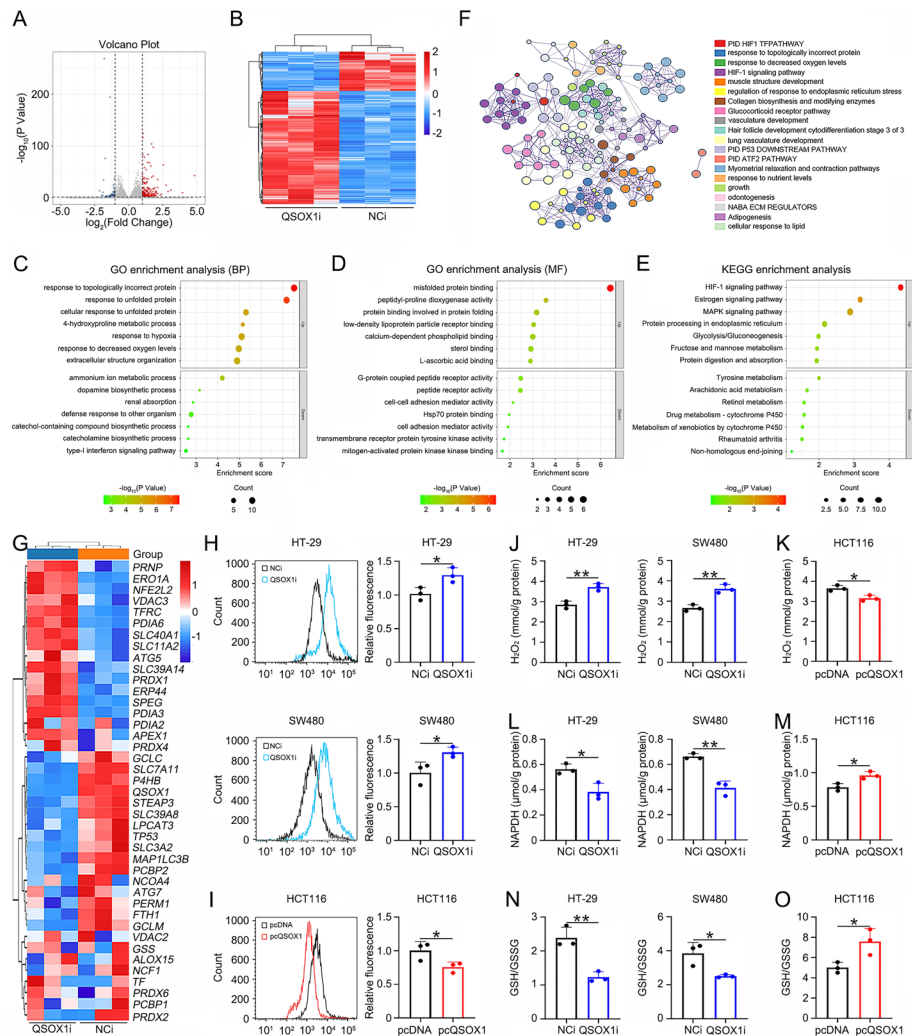


Fig. 2 QSOX1 promotes CRC progression by regulating cellular redox homeostasis. **A** Volcano plot of differentially expressed genes (DEGs) after QSOX1 depletion in HT-29 cells. **B** Heatmap of DEGs after QSOX1 depletion in HT-29 cells. **C** Gene Ontology (GO) BP analysis of DEGs after QSOX1 depletion in HT-29 cells. **D** GO MF analysis of DEGs after QSOX1 depletion in HT-29 cells. **E** Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment analysis of DEGs after QSOX1 depletion in HT-29 cells. **F** The protein-protein interaction (PPI) network of DEGs after QSOX1 depletion in HT-29 cells. **G** DEGs of cell redox homeostasis after QSOX1 depletion in HT-29 cells. **H** and **I** Flow cytometry analysis of ROS in HT-29 and SW480 cells transfected with QSOX1 siRNA **H** and HCT116 cells transfected with QSOX1 expression plasmids **I** ($n = 3$). **J** and **K** H_2O_2 contents in HT-29 and SW480 cells transfected with QSOX1 siRNA **J** and HCT116 cells transfected with QSOX1 expression plasmids **K** ($n = 3$). **L** and **M** NADPH contents in HT-29 and SW480 cells transfected with QSOX1 siRNA **L** and HCT116 cells transfected with QSOX1 expression plasmids **M** ($n = 3$). **N** and **O** GSH/GSSG ratio in HT-29 and SW480 cells transfected with QSOX1 siRNA **N** and HCT116 cells transfected with QSOX1 expression plasmids **O** ($n = 3$). Data in **H–O** are presented as mean $\pm$ SD and P -values obtained via two-tailed unpaired Student's t -test are indicated. *, $P < 0.05$; **, $P < 0.01$

correspond to changes in ROS. The results showed that QSOX1 knockdown increased H_2O_2 production in HT-29 and SW480 cells, whereas QSOX1-overexpressing HCT116 cells exhibited reduced H_2O_2 content compared with that in control cells (Fig. 2J and K). NADPH is an important metabolite for maintaining redox balance within cells. Afterward, we assayed the cellular concentration of NADPH. In contrast to ROS, NADPH concentration was significantly reduced in HT-29 and SW480 cells with QSOX1 knockdown, whereas QSOX1 overexpression increased cellular NADPH concentration in HCT116 cells (Fig. 2L and M). The ratio of reduced glutathione to oxidized glutathione

similarly reflects the redox state within cells. We found that GSH/GSSG ratio was significantly reduced in HT-29 and SW480 cells with QSOX1 under expression, whereas QSOX1 overexpression increased the GSH/GSSG ratio in HCT116 cells (Fig. 2N and O). These data suggest that QSOX1 may promote CRC progression by regulating cellular redox homeostasis.

QSOX1 exhibits a negative correlation with ferroptosis in CRC

Ferroptosis, a redox-dependent mode of regulated cell death, plays a critical role in tumor suppression. We previously demonstrated that QSOX1 regulates redox homeostasis in CRC cells, and the concurrent decrease in the GSH/GSSG ratio is a recognized hallmark of ferroptosis. Therefore, we hypothesized that QSOX1 modulates the sensitivity of CRC cells to ferroptosis. To further validate the relationship between QSOX1 and ferroptosis, MTT assays were conducted to evaluate the impact of QSOX1 knockdown on the viability of HT-29 and SW480 cells subjected to different concentrations of ferroptosis inducers (Erastin or RSL3). The results showed that the downregulation of QSOX1 expression markedly exacerbated Erastin- and RSL3-induced ferroptosis (Fig. 3A–D). In contrast, transfection of the QSOX1 plasmid into HCT116 cells enhanced their resistance to ferroptosis (Fig. 3E–H).

Ferroptosis is primarily characterized by the iron-dependent accumulation of lipid peroxides (LPO). Specifically, iron-ion-mediated Fenton reactions catalyze the conversion of endogenous H_2O_2 into cytotoxic hydroxyl radicals ($\cdot OH$), resulting in excessive LPO production on cellular membranes and subsequent triggering of ferroptosis in tumor cells. To assess this process, we measured intracellular LPO levels and malondialdehyde (MDA), a commonly used product of lipid peroxidation. We found that QSOX1 knockdown increased LPO levels in cells, whereas QSOX1 overexpression inhibited LPO production (Fig. 3I and J). Moreover, knockdown of QSOX1 expression significantly increased MDA levels in HT-29 and SW480 cells, whereas overexpression of QSOX1 in HCT116 cells decreased MDA levels (Fig. 3K and L). Furthermore, because the accumulation of redox-active Fe^{2+} is a key biochemical feature of ferroptosis, we assessed intracellular Fe^{2+} levels using both a colorimetric Fe^{2+} detection kit and the Fe^{2+} -specific fluorescent probe FerroOrange. We found that QSOX1 knockdown significantly increased intracellular Fe^{2+} levels in HT-29 and SW480 cells, whereas QSOX1 overexpression in HCT116 cells significantly reduced intracellular Fe^{2+} levels (Fig. 3M and N; Supplementary Fig. 2A and 2B). In addition, treatment with the ferroptosis-inducer Erastin markedly elevated Fe^{2+} levels in HT-29 and SW480 cells, and this effect was further enhanced by QSOX1 knockdown compared with control cells (Supplementary Fig. 3A). Conversely, QSOX1 overexpression in HCT116 cells suppressed the increase in intracellular Fe^{2+} induced by Erastin (Supplementary Fig. 3B). A similar pattern was observed following RSL3 treatment, in which QSOX1 knockdown further enhanced intracellular Fe^{2+} accumulation, whereas QSOX1 overexpression attenuated the RSL3-induced increase in Fe^{2+} levels (Supplementary Fig. 3C and 3D). Together, these findings indicate that QSOX1 restrains the accumulation of redox-active Fe^{2+} and thereby suppresses ferroptosis-associated iron dysregulation in colorectal cancer cells. To further determine whether forms of regulated cell death other than ferroptosis contribute substantially under these condition, we examined representative markers, including the apoptosis markers PARP1, caspase-3, and cleaved caspase-3, the

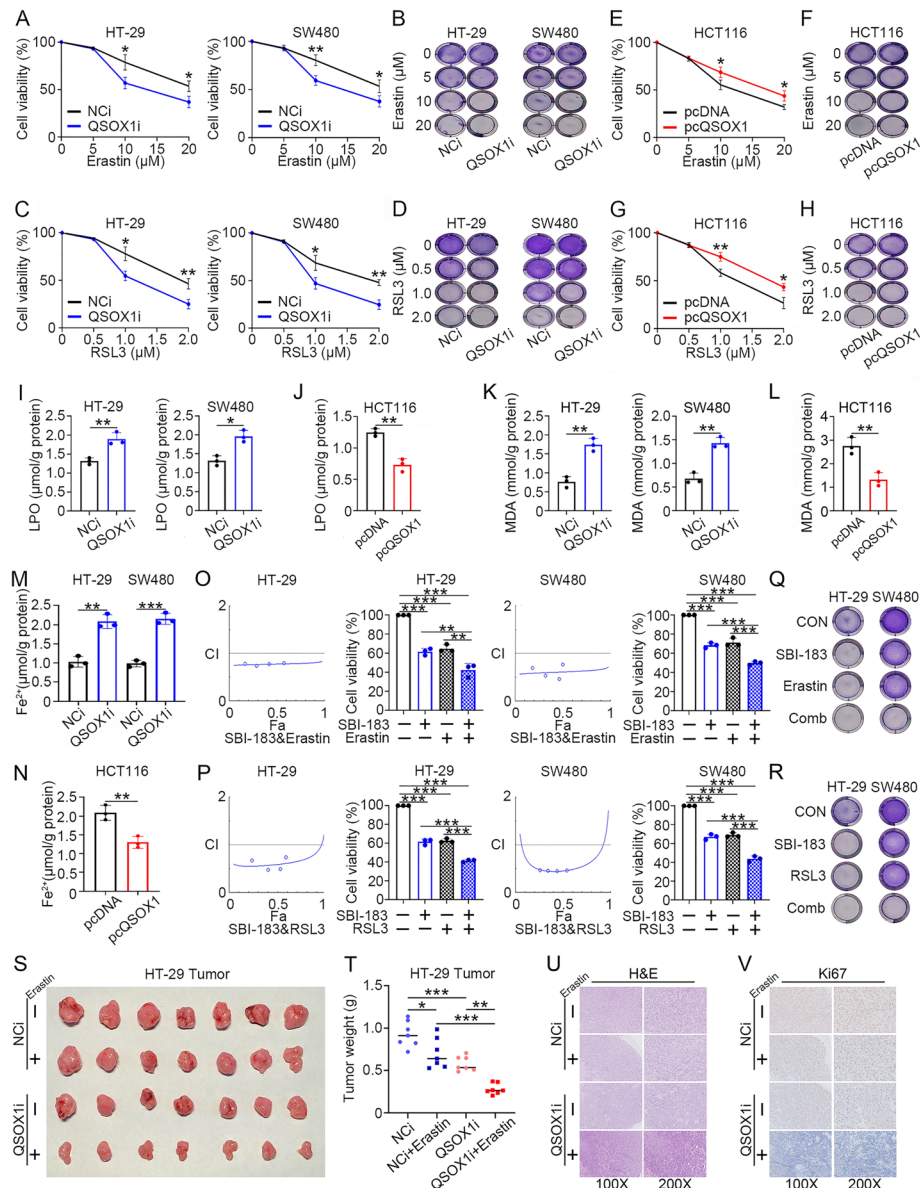


Fig. 3 QSOX1 is negatively associated with ferroptosis in CRC. **A** and **B** Cell viability was determined using the MTT assay **A** ($n = 3$) and crystalline violet staining **B** after treatment with different Erastin concentrations for 48 h in HT-29 and SW480 cells transfected with control (NCI) or QSOX1 siRNA (QSOX1i). **C** and **D** Cell viability was determined using the MTT assay **C** ($n = 3$) and crystalline violet staining **D** after treatment with different RSL3 concentrations for 48 h in HT-29 and SW480 cells transfected with control or QSOX1 siRNA. **E** and **F** Cell viability was determined using the MTT assay **E** ($n = 3$) and crystalline violet staining **F** after treatment with different Erastin concentrations for 48 h in HCT116 cells with or without transfected QSOX1 plasmid. **G** and **H** Cell viability was determined using the MTT assay **G** ($n = 3$) and crystalline violet staining **H** after treatment with different RSL3 concentrations for 48 h in HCT116 cells with or without transfected QSOX1 plasmid. **I** and **J** LPO contents in HT-29 and SW480 cells treated with siRNA targeting QSOX1 **I** and HCT116 cells transfected with QSOX1 expression plasmids **J** ($n = 3$). **K** and **L** MDA contents in HT-29 and SW480 cells treated with siRNA targeting QSOX1 **K** and HCT116 cells transfected with QSOX1 expression plasmids **L**. **M** and **N** Intracellular Fe²⁺ accumulation in HT-29 and SW480 cells treated with siRNA targeting QSOX1 **M** and HCT116 cells transfected with QSOX1 expression plasmids **N** ($n = 3$). **O** and **P** SBI-183 and Erastin **O** or RSL3 **P** combination results for combination index plot and cell viability histogram ($n = 3$). **Q** and **R** SBI-183 and Erastin **Q** or RSL3 **R** combination results for crystalline violet staining. **S** Photographs of excised tumors from four groups are shown. **T** Tumor weight was measured at the time of sacrifice in the four groups ($n = 7$). **U** and **V** Representative images of H&E **U** and Ki67 **V** staining of four groups. Data in **A–L** are presented as mean $\pm$ SD and P -values obtained via two-tailed unpaired Student's t -test are indicated. *, $P < 0.05$; **, $P < 0.01$. Data in **O**, **P**, and **T** are presented as mean $\pm$ SD and P -values obtained via one-way ANOVA with Tukey's multiple comparisons test are indicated. *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$

necroptosis marker phospho-MLKL, the autophagy-related marker LC3, and the ferroptosis marker GPX4. In HT-29 and SW480 cells, treatment with Erastin or RSL3 did not cause appreciable changes in PARP1, caspase-3, cleaved caspase-3, phospho-MLKL, or LC3, whereas GPX4 was markedly decreased (Supplementary Fig. 4A–4D). Similarly, in QSOX1-knockdown cells treated with Erastin or RSL3, PARP1, caspase-3, cleaved caspase-3, phospho-MLKL, and LC3 remained largely unchanged, whereas GPX4 was still markedly reduced (Supplementary Fig. 4A–4D). Additionally, treatment with the apoptosis inhibitor Z-VAD-FMK (Z-VAD), the necroptosis inhibitor Necrostatin-1 (Nec-1), or the autophagy inhibitor chloroquine (CQ) showed only marginal protective effects against the cell death induced by Erastin or RSL3 in QSOX1 knockdown cells (Supplementary Fig. 5A and 5B). These findings indicate that Erastin and RSL3 predominantly induce ferroptosis rather than apoptosis, necroptosis, or autophagy in CRC cells, and that QSOX1 knockdown further exacerbates this ferroptotic response. To further characterize the effect of QSOX1, we treated HT-29 and SW480 cells with SBI-183, the small molecule QSOX1 inhibitor, either alone or in combination with Erastin or RSL3. Both SBI-183 and the ferroptosis inducer inhibited cell survival, and the combination exhibited a more potent inhibitory effect (Fig. 3O and P). Combination index (CI) analysis showed that SBI-183 was synergistic with Erastin and RSL3 (Fig. 3O and P, Supplementary Fig. 6A and 6B). Crystal violet staining yielded similar results (Fig. 3Q and R). Taken together, these data indicate that QSOX1 suppresses ferroptosis in CRC cells.

Afterward, we assessed the sensitivity of control and QSOX1 knockdown CRC cells to Erastin *in vivo*. Consistent with previous results, the tumor weight of the QSOX1-knockdown (QSOX1i) group was significantly lower than that of the control (NCi) group, suggesting that QSOX1 promotes CRC cell proliferation *in vivo* (Fig. 3S and T). Furthermore, Erastin inhibited the proliferation of control tumors; however, its inhibitory effect was significantly greater in QSOX1i tumors than in control tumors (Fig. 3S and T). H&E staining revealed that the area of cell death was significantly larger in the QSOX1 knockdown + Erastin group (QSOX1i + Erastin) than in the control + Erastin group (NCi + Erastin) (Fig. 3U). Similar results were observed in Ki-67 immunohistochemical staining, where the QSOX1i + Erastin group exhibited markedly reduced Ki-67 positivity, indicating impaired cell proliferation (Fig. 3V).

These findings further confirm that QSOX1 promotes resistance to ferroptosis in CRC *in vitro* and *in vivo*. Overall, the data suggest that QSOX1 contributes to tumor progression by maintaining redox homeostasis and protecting CRC cells from ferroptosis.

Metabolomics shows that QSOX1 promotes GSH metabolism in CRC

We explored the mechanism by which QSOX1 inhibits ferroptosis in CRC. Given that RNA-seq data indicated a correlation between QSOX1 expression and CRC cell metabolism, we performed untargeted metabolomics to analyze the differences in metabolites in HT-29 cells with QSOX1 knockdown (Fig. 4A). A total of 20,138 metabolites were identified (Supplementary Table 3). A total of 72 positively charged metabolites and 95 negatively charged metabolites were differentially abundant in QSOX1i cells compared with control cells (Fig. 4B and C). Compared with control HT-29 cells, QSOX1-knockdown HT-29 cells exhibited the upregulation of 61 positively charged metabolites and downregulation of 11 positively charged metabolites (Fig. 4D and Supplementary Fig. 7).

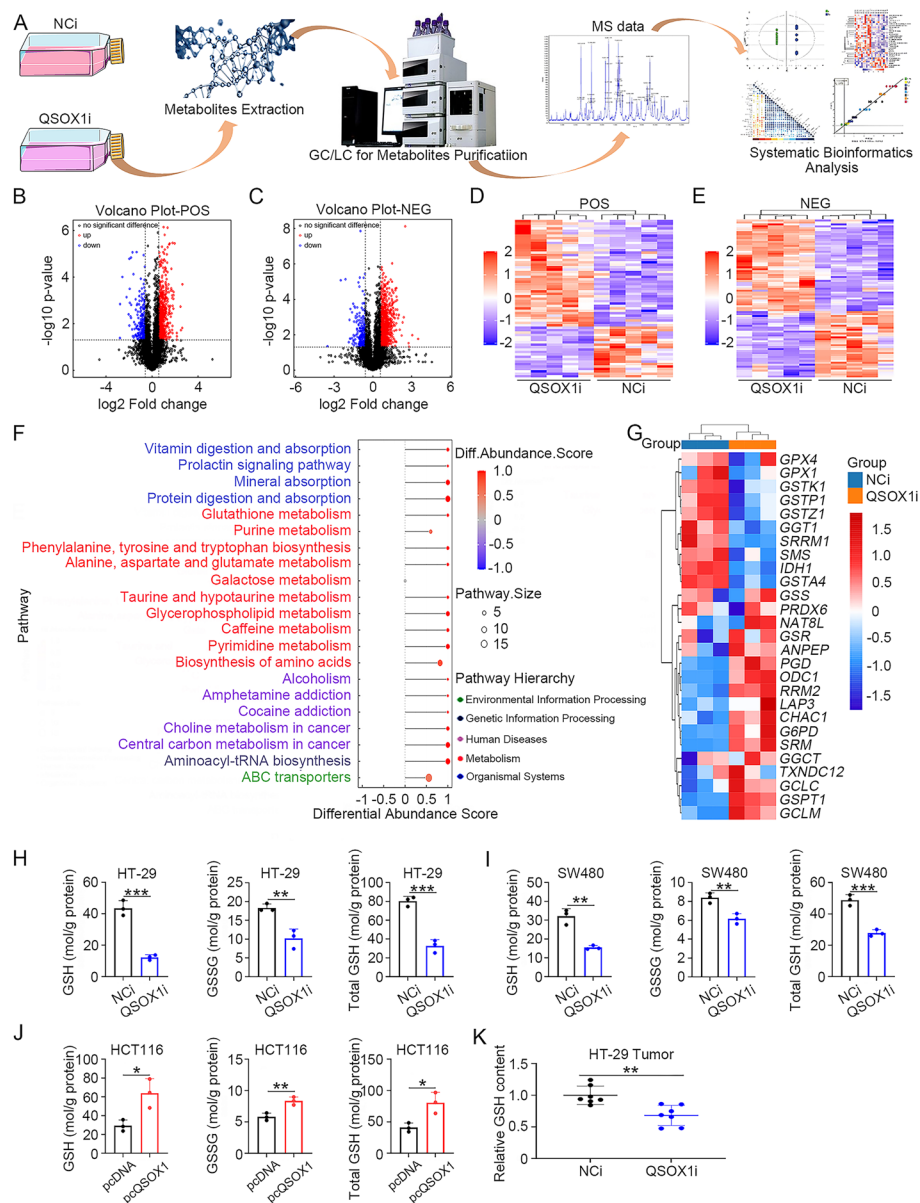


Fig. 4 Metabolomics analysis reveals that QSOX1 promotes GSH metabolism in CRC. **A** Metabolomics schematic diagram. **B** and **C** Volcano plots of differential metabolites after QSOX1 depletion in HT-29 cells in positive **B** and negative **C** ion modes. **D** and **E** Heatmaps of differential metabolites after QSOX1 depletion in HT-29 cells in positive **D** and negative **E** ion modes. **F** KEGG enrichment analysis of differential metabolites after QSOX1 depletion in HT-29 cells. **G** RNA-seq DEGs of GSH metabolism after QSOX1 depletion in HT-29 cells. **H–J** GSH, GSSG, and total cellular glutathione concentrations in HT-29 **H** and SW480 cells **I** transfected with siRNA targeting QSOX1 and HCT116 cells transfected with QSOX1 expression plasmids **J** ($n = 3$). **K** Relative content of GSH in mouse tumor tissues ($n = 7$). Data in **H–K** are presented as mean $\pm$ SD and P -values obtained via two-tailed unpaired Student's t -test are indicated. **, $P < 0.01$; ***, $P < 0.001$

Additionally, 78 negatively charged metabolites were upregulated, whereas 17 were downregulated (Fig. 4E and Supplementary Fig. 8).

KEGG pathway analysis of the differentially abundant metabolites indicated their participation in pathways closely related to tumor progression, such as GSH metabolism, amino acid biosynthesis and metabolism, and one-carbon metabolism in cancer (Fig. 4F). GSH metabolism is a major cellular antioxidant pathway and plays a crucial

role in cancer resistance to ferroptosis. Our results showed that the expression of genes involved in GSH synthesis was downregulated in QSOX1-knockdown HT-29 cells based on RNA-seq data, whereas the expression of genes involved in GSH transferases was upregulated (Fig. 4G).

These findings suggest that QSOX1 may regulate glutathione homeostasis in CRC cells. We therefore measured intracellular levels of GSH, GSSG, and total glutathione. The results showed that the GSH, GSSG, and total cellular glutathione concentration in QSOX1 knockdown HT-29 and SW480 cells were significantly reduced compared with the concentrations in control cells (Fig. 4H and I). Moreover, QSOX1 overexpression in HCT116 cells significantly increased the concentration of GSH, GSSG, and total cellular glutathione (Fig. 4J). Furthermore, in vivo experiment demonstrated that the total cellular GSH concentration in tumor tissues of the QSOX1i group was markedly reduced compared with the NCi group (Fig. 4K). To determine whether the increased ferroptosis sensitivity caused by QSOX1 downregulation is mediated by glutathione depletion, HT-29 and SW480 cells with QSOX1 knockdown were treated with Erastin or RSL3 in the presence or absence of exogenous GSH. The results showed that GSH supplementation markedly attenuated the enhanced ferroptotic sensitivity induced by QSOX1 downregulation (Supplementary Fig. 9A and 9B). Together, these findings suggest that QSOX1 maintains glutathione homeostasis in CRC cells, and that glutathione depletion contributes to the increased susceptibility to ferroptosis caused by QSOX1 loss.

QSOX1 increases the resistance of CRC cells to ferroptosis through SLC7A11

SLC7A11 is a key regulator of ferroptosis and enhances GSH synthesis by facilitating the uptake of cystine into cells [20]. SLC7A11 inhibitor Erastin reduces cystine uptake and intracellular GSH synthesis, thereby triggering ferroptosis [4]. Our data indicate that QSOX1 promotes the GSH synthesis pathway in CRC cells. We further investigated whether the promotion of ferroptosis resistance by QSOX1 in CRC depends on SLC7A11. The results showed that SLC7A11 overexpression rescued the increased sensitivity to Erastin- and RSL3-induced ferroptosis and MDA levels caused by QSOX1 knockdown in HT-29 and SW480 cells (Fig. 5A and B, Supplementary Fig. 10A). Similarly, SLC7A11 overexpression inhibited the activity of SBI-183 in combination with Erastin and RSL3 (Supplementary Fig. 11A and 11B). Moreover, SLC7A11 knockdown in HCT116 cells reversed the resistance to Erastin- and RSL3-induced ferroptosis conferred by QSOX1 overexpression (Fig. 5C and D). Similarly, SLC7A11 knockdown also reversed the QSOX1-mediated reduction in MDA levels (Supplementary Fig. 10B).

These data suggest that QSOX1-mediated regulation of ferroptosis in CRC depends on SLC7A11. Furthermore, QSOX1 possesses molecular chaperone activity and may enhance the function of substrate proteins through interaction. Immunoprecipitation assays showed that QSOX1 interacts with SLC7A11 in CRC cells (Fig. 5E). Immunofluorescence staining confirmed that QSOX1 colocalizes with SLC7A11 (Fig. 5F). Furthermore, immunofluorescence staining shows that QSOX1 knockdown inhibited the membrane localization of SLC7A11, whereas QSOX1 overexpression increased SLC7A11 membrane localization (Fig. 5G and H). SLC7A11 imports cystine into the cell and exports glutamate outside the cell by associating with SLC3A2 to form the System Xc⁻ antiporter complex. QSOX1 knockdown significantly inhibited cystine uptake in HT-29 and SW480 cells, and this effect was rescued by SLC7A11 overexpression

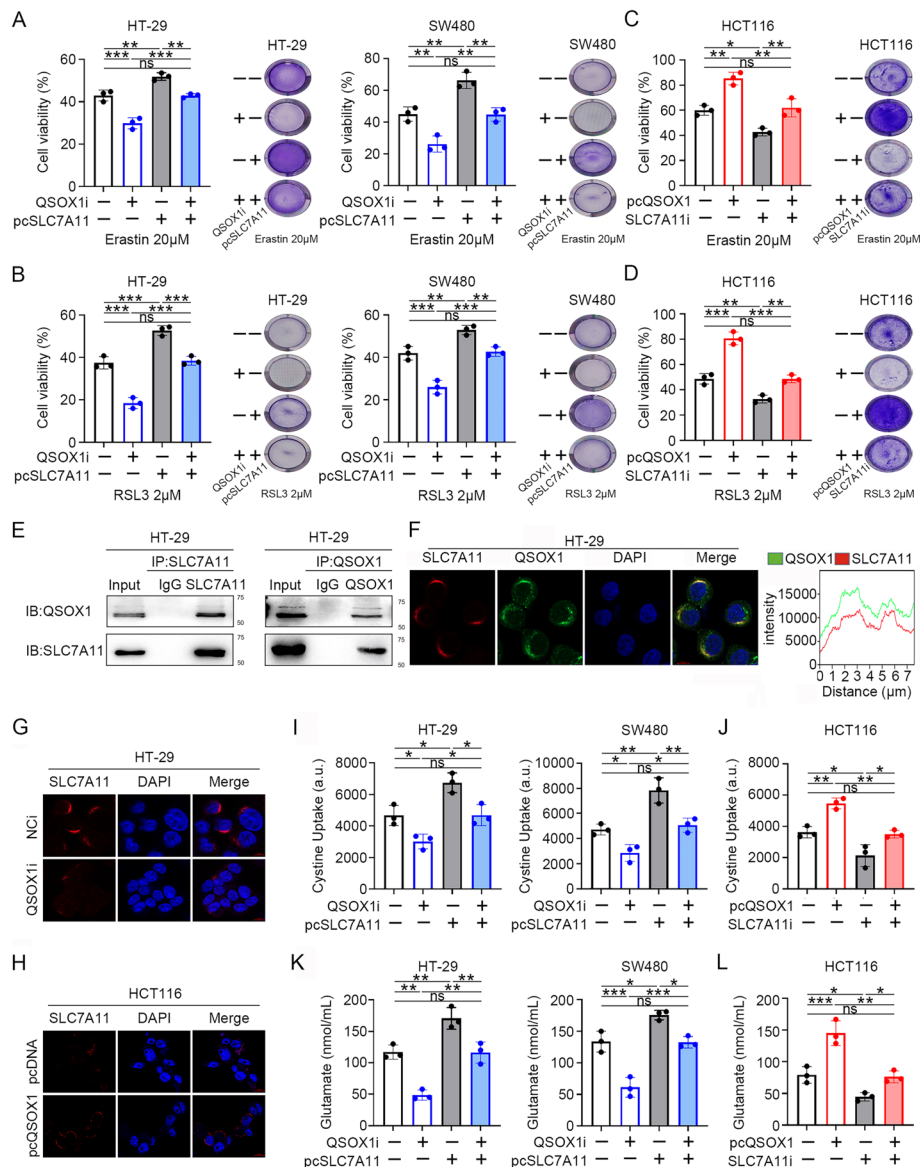


Fig. 5 QSOX1 promotes CRC resistance to ferroptosis through SLC7A11. **A** and **B** The cotransfection assays showed that overexpression of SLC7A11 in HT-29 and SW480 cells reversed QSOX1 knockdown-induced sensitization to Erastin **A** and RSL3 **B** ($n = 3$). **C** and **D** The cotransfection assays showed that knockdown of SLC7A11 in HCT116 cells reversed QSOX1 overexpression-induced resistance to Erastin **C** and RSL3 **D** ($n = 3$). **E** Co-IP analysis of the interaction between QSOX1 and SLC7A11 in HT-29 cells. **F** Confocal analysis of the colocalization of QSOX1 (red) and SLC7A11 (green) in HT-29 cells. Intensity profiles of each line were quantified by ZEN2 software. **G** HT-29 cells treated with siRNA targeting QSOX1 were stained with SLC7A11 (red) and DAPI (blue). **H** HCT116 cells treated with QSOX1 expression plasmids were stained with SLC7A11 (red) and DAPI (blue). **I** and **J** Effect of SLC7A11 on QSOX1-mediated cystine uptake. Cellular cystine uptake was assayed after cotransfection of HT-29 and SW480 cells with QSOX1 siRNA and SLC7A11 plasmid **I** or after transfection of HCT116 cells with QSOX1 expression plasmid and SLC7A11 siRNA **J** ($n = 3$). **K** and **L** Effect of SLC7A11 on QSOX1-mediated glutamate release. Cellular glutamate release was assayed after cotransfection of HT-29 and SW480 cells with QSOX1 siRNA and SLC7A11 plasmid **K** or after transfection of HCT116 cells with QSOX1 expression plasmid and SLC7A11 siRNA **L** ($n = 3$). Data in **A–D** and **I–L** are presented as mean $\pm$ SD and P -values obtained via one-way ANOVA with Tukey's multiple comparisons test are indicated. ns, not significant; *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$

(Fig. 5I). Moreover, QSOX1 overexpression in HCT116 cells significantly increased cystine uptake, and this effect was reversed by SLC7A11 knockdown (Fig. 5J). Similarly, QSOX1 knockdown significantly inhibited glutamate release, whereas SLC7A11 overexpression suppressed this inhibition (Fig. 5K). Furthermore, SLC7A11 knockdown reversed the increase in glutamate efflux caused by QSOX1 overexpression (Fig. 5L). These findings suggest that QSOX1 promotes SLC7A11 function through protein–protein interaction, thereby conferring ferroptosis resistance.

QSOX1 promotes SLC7A11 activity through its second thioredoxin motif in the ERV/ALR domain

QSOX1 is a sulfhydryl oxidase that promotes the formation of intramolecular or intermolecular disulfide bonds at cysteine residues of substrate proteins [21]. Therefore, we used nonreducing PAGE to investigate the effect of QSOX1 on cysteine redox modification of SLC7A11. Redox modification in SLC7A11 was significantly decreased in HT-29 cells with low QSOX1 expression, whereas QSOX1 overexpression in HCT116 cells significantly promoted the redox modification of SLC7A11 (Fig. 6A and B, Supplementary Fig. 12A and 12B). Conversely, SLC7A11 displayed a significantly more redox modification following QSOX1 overexpression in HCT116 cells; this modification could be inhibited by the disulfide bond reducer TCEP (Fig. 6C, Supplementary Fig. 12C). These results suggest that QSOX1 promotes the redox modification of SLC7A11.

CXXC motifs, which mediate cysteine oxidation and disulfide bond formation, are important thioredoxin motifs. QSOX1 contains three CXXC motifs: C70–C73 located in the Trx1 domain and C449–C452, and C509–C512 located in the ERV/ALR domain (Fig. 6D). We mutated the cysteine residues in these three CXXC motifs to alanine and subsequently transfected cells to assess the redox modification of SLC7A11. The results showed that the C449A–C452A QSOX1 mutant exhibited a drastically reduced interaction with SLC7A11 and lost the ability to promote SLC7A11 redox modification (Fig. 6E and F, Supplementary Fig. 12D).

Functional assays of SLC7A11 showed that transfection of QSOX1 C449A–C452A failed to promote cystine uptake and glutamate release in HCT116 cells (Fig. 6G and H). MTT assay results indicated that QSOX1 C449A–C452A did not promote the resistance of CRC cells to Erastin and RSL3 (Fig. 6I). Similar results were obtained from crystal violet staining (Fig. 6J). These data suggest that QSOX1 promotes cysteine-oxidative modification and function of SLC7A11 via the C449–C452 motif in the ERV/ALR domain.

Cysteine 158 of SLC7A11 is crucial for the regulation of SLC7A11 activity

We next investigated which cysteine residue of SLC7A11 is modified by QSOX1. SLC7A11 contains seven cysteine residues; we mutated each of these cysteines to alanine individually and transfected them into HCT116 cells along with QSOX1 (Fig. 7A). QSOX1 failed to promote the redox modification of SLC7A11 C158A, and SLC7A11 C158A exhibited a loss of interaction with QSOX1 (Fig. 7B and C, Supplementary Fig. 12E). These findings suggest that Cys158 of SLC7A11 is crucial for QSOX1-mediated redox modification.

Structural studies have shown that Cys158 of SLC7A11 can form a disulfide bond with Cys210 of SLC3A2 [22]. We found that QSOX1 overexpression significantly enhanced the interaction between SLC7A11 and SLC3A2 (Fig. 7D). The interaction

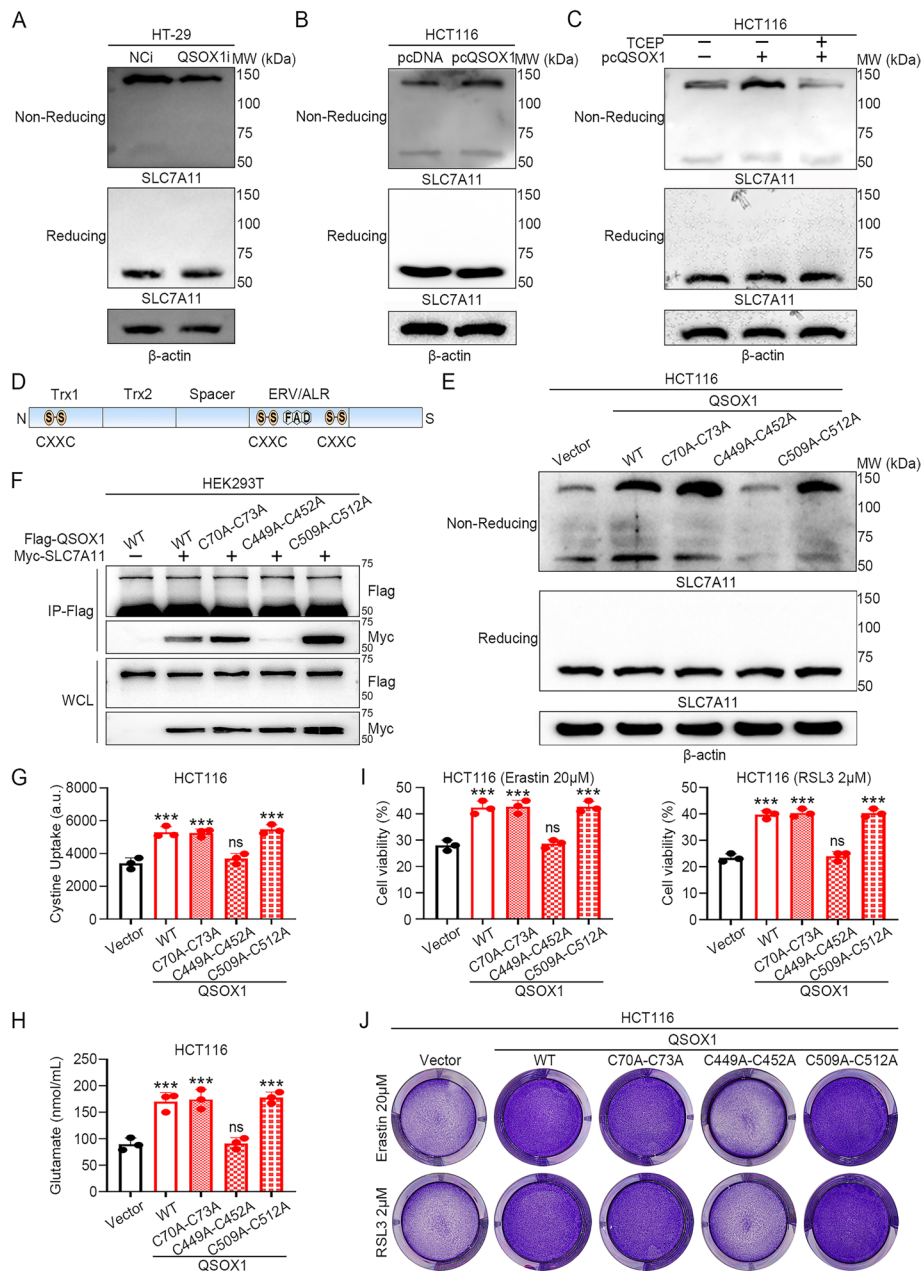


Fig. 6 QSOX1 promotes SLC7A11 activity through its second thioredoxin motif within the ERV/ALR domain. **A** Nonreduced and reduced immunoblots of the lysates of HT-29 cells transfected with QSOX1 siRNA. **B** Nonreduced and reduced immunoblots of the lysates of HCT116 cells transfected with QSOX1 overexpression plasmid. **C** Non-reduced and reduced immunoblots of the lysates of HCT116 cells transfected with plasmids expressing QSOX1 and treatment with or without TCEP. **D** Schematic diagram of human QSOX1 structural domains. **E** Co-IP analysis of the interaction between SLC7A11 and QSOX1 or QSOX1 mutants in HEK293T cells transfected with the plasmids expressing Myc-SLC7A11, with Flag-QSOX1, Flag-QSOX1 C70A-C73A, Flag-QSOX1 C449A-C452A, or Flag-QSOX1 C509A-C512A. **F** Nonreduced and reduced immunoblots of the lysates of HCT116 cells transfected with plasmids expressing SLC7A11, with QSOX1, QSOX1 C70A-C73A, QSOX1 C449A-C452A, or QSOX1 C509A-C512A. **G** and **H** Effect of QSOX1 functional mutants in cellular cystine uptake and glutamate release. Cellular cystine uptake **G** and glutamate release **H** were measured in HCT116 cells transfected with plasmids expressing QSOX1, QSOX1 C70A-C73A, QSOX1 C449A-C452A, or QSOX1 C509A-C512A ($n = 3$). **I** and **J** Cell viability was determined using the MTT assay **I** and crystalline violet staining **J** after treatment with Erastin or RSL3 for 48 h in HCT116 cells transfected with plasmids expressing QSOX1, QSOX1 C70A-C73A, QSOX1 C449A-C452A, or QSOX1 C509A-C512A ($n = 3$). Data in **G–I** are presented as mean $\pm$ SD and P -values obtained via one-way ANOVA with Dunnett's multiple comparisons test are indicated. ns, not significant; ***, $P < 0.001$

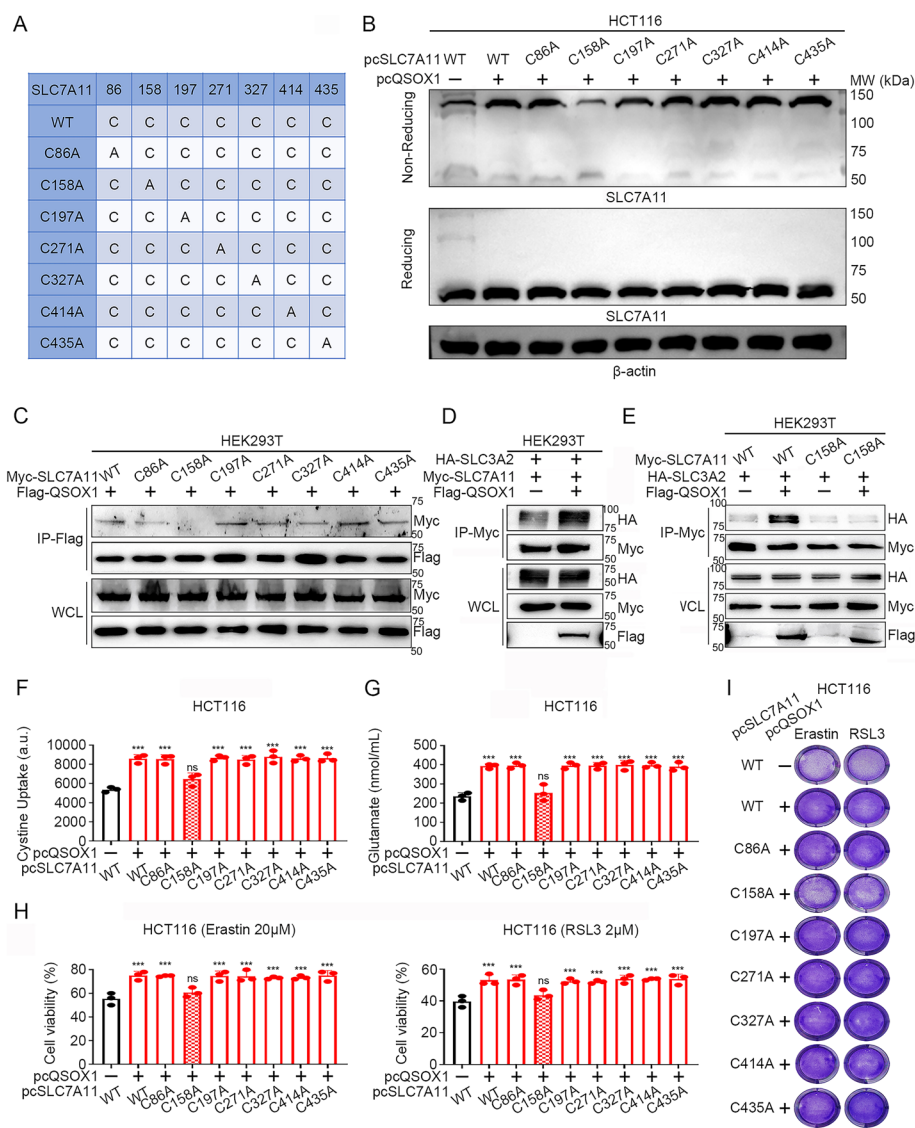


Fig. 7 Cysteine 158 of SLC7A11 is crucial for regulating SLC7A11 activity. **A** Schematic diagram of human SLC7A11 and its mutants. **B** Nonreduced and reduced immunoblots of the lysates of HCT116 cells transfected with plasmids expressing QSOX1 and wildtype or mutated SLC7A11. **C** Co-IP analysis of the interaction between QSOX1 and SLC7A11 or SLC7A11 mutants in HEK293T cells. **D** Co-IP analysis of the interaction between SLC7A11 and SLC3A2 in HEK293T cells with (+) or without (-) transfected QSOX1. **E** Co-IP analysis of the interaction between SLC3A2 and SLC7A11 or SLC7A11 C158A mutant in HEK293T cells with or without transfected QSOX1. **F** and **G** Effect of SLC7A11 functional mutants in cellular cystine uptake and glutamate release. Cellular cystine uptake **F** and glutamate release **G** were measured in HCT116 cells transfected with plasmids expressing QSOX1 and wildtype or mutated SLC7A11 ($n = 3$). **H** and **I** Cell viability was determined using the MTT assay **H** ($n = 3$) and crystalline violet staining **I** after treatment with Erastin or RSL3 for 48 h in HCT116 cells transfected with plasmids expressing QSOX1 wildtype or mutated SLC7A11. Data in **F–H** are presented as mean $\pm$ SD and P -values obtained via one-way ANOVA with Dunnett's multiple comparisons test are indicated. ns, not significant; ***, $P < 0.001$

between SLC7A11 C158A and SLC3A2 was significantly reduced, and its enhancement by QSOX1 was inhibited (Fig. 7E). Functional assays of SLC7A11 revealed that cystine uptake and glutamate release activities were significantly diminished in the SLC7A11 C158A mutant (Fig. 7F and G). Cell viability assays indicated that the SLC7A11 C158A mutant did not synergize with QSOX1 to promote the resistance of HCT116 cells to Erastin and RSL3 (Fig. 7H). Similar results were obtained from crystal violet staining

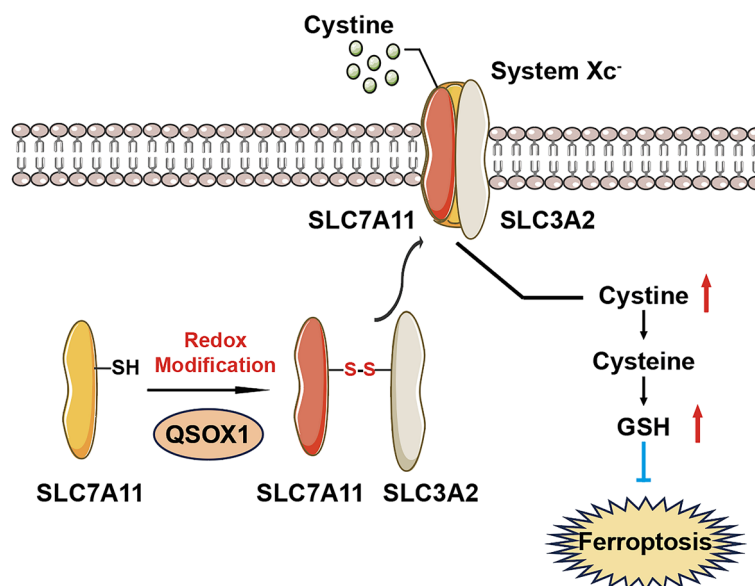


Fig. 8 The proposed model of QSOX1 induces CRC cell ferroptosis resistance. QSOX1 enhances the interaction between SLC7A11 and SLC3A2 and promotes SLC7A11 activity through redox-dependent post-translational modification, increasing cellular cystine uptake and GSH homeostasis, thereby promoting CRC cell resistance to ferroptosis

(Fig. 7I). These findings suggest that Cys158 is a critical cysteine residue involved in the regulation of SLC7A11 activity by QSOX1.

Discussion

This study elucidated the critical role of QSOX1 in modulating ferroptosis resistance and redox homeostasis in CRC, underscoring its potential as a therapeutic target in ferroptosis-based cancer therapies. QSOX1, a sulfhydryl oxidase, regulated the redox environment within CRC cells by promoting thiol oxidation and enhancing the function of SLC7A11, a key regulator of ferroptosis. Our findings indicate that QSOX1 promotes CRC progression by maintaining redox homeostasis and inhibiting ferroptosis (Fig. 8).

Ferroptosis has emerged as a distinct form of cell death essential for regulating tumor growth, therapy response, and immune sensitivity [23–25]. Its execution is tightly linked to metabolic pathways governing iron homeostasis, lipid metabolism, and antioxidant defense. The ability of cancer cells to resist ferroptosis is increasingly recognized as a critical mechanism underlying tumor progression and treatment resistance. In this context, our results revealed that QSOX1 plays a crucial role in redox regulation, which is critical for the resistance of CRC cells to ferroptosis. Specifically, QSOX1 knockdown in CRC cell lines led to increased ROS levels, lipid peroxidation, intracellular Fe²⁺ levels, and enhanced cellular sensitivity to ferroptosis, whereas QSOX1 overexpression resulted in decreased sensitivity to ferroptosis. These findings corroborate those of recent studies demonstrating the importance of redox-regulating enzymes, including GPX4 [26, 27], FSP1 [28], and DHODH [29], in CRC ferroptosis suppression. Our study expands this network by identifying QSOX1 as a redox-modulating enzyme that cooperates with these established ferroptosis regulators to sustain CRC cell viability.

Furthermore, the transcriptomic and metabolomic data from this study indicate that QSOX1 knockdown negatively affects key metabolic pathways, including those involved

in GSH synthesis and amino acid metabolism. These metabolic alterations play a central role in neutralizing lipid hydroperoxides and limiting ferroptosis, and GSH synthesis depends on the availability of cysteine, which is imported via the System Xc⁻ antiporter [30]. The observed decrease in GSH levels following QSOX1 knockdown suggests a mechanistic link between QSOX1 activity and cystine transport, most likely mediated through SLC7A11. Our study found that QSOX1 expression is positively correlated with GSH levels in CRC cells, suggesting that GSH metabolism is a central pathway through which QSOX1 promotes CRC cell resistance to ferroptosis. For instance, in hepatocellular carcinoma, aberrant upregulation of the transsulfuration pathway compensates for cysteine depletion to maintain GSH levels and limit ferroptosis [31]. Similarly, in lung adenocarcinoma, NRF2-driven SLC7A11 expression or SPTBN2 facilitating SLC7A11 membrane trafficking enables GSH-dependent ferroptosis evasion [32, 33]. Our study adds a new layer to this regulatory network by identifying QSOX1 as a facilitator of cystine-dependent GSH metabolism.

An important mechanism through which QSOX1 affects redox regulation in CRC is its interaction with SLC7A11, a central regulator of ferroptosis. SLC7A11, a component of the System Xc⁻ antiporter complex, facilitates cystine uptake, which is subsequently converted to GSH. Our findings demonstrate that QSOX1 interacts and promotes a redox post-translational modification (PTM) of SLC7A11, thereby enhancing its transport activity. This finding is consistent with our multi-omics data. Moreover, this finding aligns with recent findings showing that SLC7A11 upregulation is a common mechanism underlying ferroptosis resistance in various cancers, including CRC [34–37]. Notably, we demonstrate that QSOX1-mediated resistance to ferroptosis is dependent on SLC7A11, as SLC7A11 overexpression rescued the increased sensitivity to ferroptosis caused by QSOX1 knockdown. These findings suggest that the redox state of SLC7A11 is a key determinant of its functional activity and highlight QSOX1 as a noncanonical modulator of this transporter.

The functional role of PTMs in modulating SLC7A11 activity and ferroptosis sensitivity has attracted increasing attention. For example, SOCS1 and SOCS2 promote ferroptosis by mediating the ubiquitination and degradation of SLC7A11 [38, 39], whereas USP18, a deubiquitinating enzyme, stabilizes SLC7A11, thereby inhibiting ferroptosis in breast cancer cells [40]. Other PTMs such as phosphorylation, O-GlcNAcylation, and UFMylation can regulate SLC7A11 activity [41–43]. However, although SLC7A11 is redox-regulated, oxidative modification of SLC7A11 has not been reported. We suggest that QSOX1 interacts with SLC7A11 to promote the redox modification of Cys158 in SLC7A11, further highlighting the specificity of QSOX1-mediated ferroptosis regulation. The cystine transport activity of SLC7A11 with Cys158 mutations is significantly reduced in CRC cells. Besides, SLC7A11 with Cys158 mutations does not synergistically promote ferroptosis resistance in CRC cells in the presence of QSOX1, suggesting that Cys158 is a key amino acid for SLC7A11 functional activity. Structural studies indicate that Cys158 participates in the formation of disulfide bonds with SLC3A2. Furthermore, our results showed reduced interaction between Cys158 of SLC7A11 and SLC3A2, emphasizing the importance of cysteine oxidation in regulating System Xc⁻ complex activity. SLC3A2 is a component of heteromeric amino acid transporters (HATs) and plays an essential role in amino acid transport [44]. The cysteine residues within the extracellular domain of SLC3A2 can form disulfide bonds with six L-type amino acid

transporters (LATs) from the SLC7 family—including SLC7A5, SLC7A6, SLC7A7, SLC7A8, SLC7A10, and SLC7A11—thereby assembling functional HAT complexes [22, 45]. These disulfide bonds between SLC3A2 and LATs are critical for amino acid transport activity [46]. In the present study, we demonstrated that QSOX1 functions as the essential sulfhydryl oxidase responsible for catalyzing disulfide bond formation between SLC3A2 and SLC7A11. Whether QSOX1 also facilitates disulfide bond formation between SLC3A2 and other LAT family members remains to be determined.

Importantly, our *in vivo* experiments validated the protumorigenic role of QSOX1 and its contribution to ferroptosis resistance. CRC xenografts with QSOX1 knockdown exhibited reduced tumor growth and were significantly more sensitive to Erastin treatment, indicating that QSOX1 suppression restores ferroptosis vulnerability *in vivo*. This observation is consistent with recent evidence that ferroptosis reactivation in resistant tumors can effectively suppress tumor progression. Notably, small-molecule QSOX1 inhibitors have recently been developed [47], and their application in combination with ferroptosis inducers warrants further investigation. Therefore, pharmacological inhibition of QSOX1 could represent a promising strategy to lower the threshold for ferroptosis induction in CRC. Our study demonstrated that combination therapy with QSOX1 inhibitors and ferroptosis inducers such as Erastin or RSL3 yielded synergistic anti-tumor effects in CRC cells. Moreover, the identification of Cys158 as a redox-sensitive hotspot in SLC7A11 offers a promising avenue for the development of ferroptosis-sensitizing therapeutic strategies targeting CRC.

Conclusions

Our study provides compelling evidence that QSOX1 regulates ferroptosis resistance in CRC by modulating SLC7A11 activity and maintaining cellular redox homeostasis. Given the established relationship between ferroptosis resistance and poor prognosis in CRC, targeting QSOX1 or its downstream pathways represents a promising therapeutic strategy for enhancing the efficacy of ferroptosis-based therapies. Future studies will be necessary to further elucidate the precise mechanisms by which QSOX1 regulates ferroptosis and evaluate the therapeutic potential of QSOX1 inhibition in clinical settings.

Abbreviations

CRC	Colorectal cancer
ROS	Reactive oxygen species
LPO	Lipid peroxides
LOXs	Lipoxygenases
SLC7A11	Solute carrier family 7 member 11
SLC3A2	Solute carrier family 3 member 2
GSH	Glutathione
GPX4	Glutathione peroxidase 4
QSOX1	Quiescin sulfhydryl oxidase 1
H&E	Hematoxylin and eosin
CI	Combination index
DEGs	Differentially expressed genes
ELISA	Enzyme-linked immunosorbent assay
GO	Gene ontology
BP	Biological process
MF	Molecular functions
KEGG	Kyoto encyclopedia for genes and genomes
PPI	Protein–protein interaction
H ₂ O ₂	Hydrogen peroxide
PTM	Post-translational modification
HATs	Heteromeric amino acid transporters
LATs	L-type amino acid transporters

Supplementary Information

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Additional file 1.

Additional file 2.

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Author contributions

M. J. conceived and designed the study and wrote the manuscript. H. C., Y. L., J. G., X.W., Q. L., Z. L. Z. M., and C. W. performed the experiments, analyzed and interpreted the results. Y. W. performed the experiments, analyzed and interpreted the results and wrote the manuscript. All authors reviewed the manuscript.

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Data availability

The RNA-Seq data generated in this study have been deposited in the NCBI Sequence Read Archive (SRA) database under accession code PRJNA1280752. All data are available in the main text and/or the supplementary materials. Additional data related to this paper may be requested from the authors.

Declarations

Ethics approval and consent to participate

All CRC cases were approved by the Medical Ethics Committee of the First Affiliated Hospital of Shandong First Medical University (YXLL-KY-2023-002), with informed consent provided by all patients. All animal experiments were approved by the Institutional Animal Care and Use Committee of the First Affiliated Hospital of Shandong First Medical University (2024010901).

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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References

1. Siegel RL, Wagle NS, Cercak A, Smith RA, Jemal A. Colorectal cancer statistics, 2023. *CA Cancer J Clin.* 2023;73(3):233–54.
2. Siegel RL, Kratzer TB, Giaquinto AN, Sung H, Jemal A. Cancer statistics, 2025. *CA Cancer J Clin.* 2025;75(1):10–45.
3. Yan H, Talty R, Johnson CH. Targeting ferroptosis to treat colorectal cancer. *Trends Cell Biol.* 2023;33(3):185–8.
4. Dixon SJ, Lemberg KM, Lamprecht MR, Skouta R, Zaitsev EM, Gleason CE, et al. Ferroptosis: an iron-dependent form of nonapoptotic cell death. *Cell.* 2012;149(5):1060–72.
5. Yang WS, Kim KJ, Gaschler MM, Patel M, Shchepinov MS, Stockwell BR. Peroxidation of polyunsaturated fatty acids by lipoxygenases drives ferroptosis. *Proc Natl Acad Sci U S A.* 2016;113(34):E4966–75.
6. Koppula P, Zhang Y, Zhuang L, Gan B. Amino acid transporter SLC7A11/xCT at the crossroads of regulating redox homeostasis and nutrient dependency of cancer. *Cancer Commun.* 2018;38(1):12.

7. Yang WS, SriRamaratnam R, Welsch ME, Shimada K, Skouta R, Viswanathan VS, et al. Regulation of ferroptotic cancer cell death by GPX4. *Cell*. 2014;156(1–2):317–31.
8. Sun R, Lin Z, Wang X, Liu L, Huo M, Zhang R, et al. AADAC protects colorectal cancer liver colonization from ferroptosis through SLC7A11-dependent inhibition of lipid peroxidation. *J Exp Clin Cancer Res*. 2022;41(1):284.
9. Lei G, Zhuang L, Gan B. Targeting ferroptosis as a vulnerability in cancer. *Nat Rev Cancer*. 2022;22(7):381–96.
10. Wang L, Wang CC. Oxidative protein folding fidelity and redox in the endoplasmic reticulum. *Trends Biochem Sci*. 2023;48(1):40–52.
11. Lake DF, Faigel DO. The emerging role of QSOX1 in cancer. *Antioxid Redox Signal*. 2014;21(3):485–96.
12. Reznik N, Fass D. Disulfide bond formation and redox regulation in the Golgi apparatus. *FEBS Lett*. 2022;596(22):2859–72.
13. Ilani T, Alon A, Grossman I, Horowitz B, Kartvelishvili E, Cohen SR, et al. A secreted disulfide catalyst controls extracellular matrix composition and function. *Science*. 2013;341(6141):74–6.
14. Millar-Haskell CS, Sperduto JL, Slater JH, Thorpe C, Gleghorn JP. Secretion of the disulfide bond generating catalyst QSOX1 from pancreatic tumor cells into the extracellular matrix: association with extracellular vesicles and matrix proteins. *J Extracell Biol*. 2022;1(7).
15. Wei JR, Zhang B, Zhang Y, Chen WM, Zhang XP, Zeng TT, et al. QSOX1 facilitates dormant esophageal cancer stem cells to evade immune elimination via PD-L1 upregulation and CD8 T cell exclusion. *Proc Natl Acad Sci U S A*. 2024;121(44):e2407506121.
16. Bartha A, Gyorffy B. TNMplot.com: a web tool for the comparison of gene expression in normal, tumor and metastatic tissues. *Int J Mol Sci*. 2021. <https://doi.org/10.3390/ijms22052622>.
17. Gyorffy B. Integrated analysis of public datasets for the discovery and validation of survival-associated genes in solid tumors. *Innovation (Camb)*. 2024;5(3):100625.
18. Heckler EJ, Rancy PC, Kodali VK, Thorpe C. Generating disulfides with the Quiescin-sulfhydryl oxidases. *Biochim Biophys Acta*. 2008;1783(4):567–77.
19. Sun J, Zhou C, Zhao Y, Zhang X, Chen W, Zhou Q, et al. Quiescin sulfhydryl oxidase 1 promotes sorafenib-induced ferroptosis in hepatocellular carcinoma by driving EGFR endosomal trafficking and inhibiting NRF2 activation. *Redox Biol*. 2021;41:101942.
20. Koppula P, Zhuang L, Gan B. Cystine transporter SLC7A11/xCT in cancer: ferroptosis, nutrient dependency, and cancer therapy. *Protein Cell*. 2021;12(8):599–620.
21. Chakravarthi S, Jessop CE, Willer M, Stirling CJ, Bulleid NJ. Intracellular catalysis of disulfide bond formation by the human sulfhydryl oxidase, QSOX1. *Biochem J*. 2007;404(3):403–11.
22. Yan R, Xie E, Li Y, Li J, Zhang Y, Chi X, et al. The structure of erastin-bound xCT-4F2hc complex reveals molecular mechanisms underlying erastin-induced ferroptosis. *Cell Res*. 2022;32(7):687–90.
23. Wang X, Ren X, Lin X, Li Q, Zhang Y, Deng J, et al. Recent progress of ferroptosis in cancers and drug discovery. *Asian J Pharm Sci*. 2024;19(4):100939.
24. Zhou Q, Meng Y, Le J, Sun Y, Dian Y, Yao L, et al. Ferroptosis: mechanisms and therapeutic targets. *MedComm*. 2024;5(12):e70010.
25. Xu Y, Ge M, Xu Y, Yin K. Ferroptosis: a novel perspective on tumor immunotherapy. *Front Immunol*. 2025;16:1524711.
26. Zhang X, Wang Z, He Y, Wang K, Xiang C, Liu Y, et al. ARID1A loss enhances sensitivity to c-MET inhibition by dual targeting of GPX4 and iron homeostasis, inducing ferroptosis. *Cell Death Differ*. 2025. <https://doi.org/10.1038/s41418-025-01510-x>.
27. Zhou X, Zhao Y, Huang S, Shu H, Zhang Y, Yang H, et al. TRIM32 promotes neuronal ferroptosis by enhancing K63-linked ubiquitination and subsequent p62-selective autophagic degradation of GPX4. *Int J Biol Sci*. 2025;21(3):1259–74.
28. Zheng X, Wang Q, Zhou Y, Zhang D, Geng Y, Hu W, et al. N-acetyltransferase 10 promotes colon cancer progression by inhibiting ferroptosis through N4-acetylation and stabilization of ferroptosis suppressor protein 1 (FSP1) mRNA. *Cancer Commun*. 2022;42(12):1347–66.
29. Dong S, Zhang M, Cheng Z, Zhang X, Liang W, Li S, et al. Redistribution of defective mitochondria-mediated dihydroorotate dehydrogenase imparts 5-fluorouracil resistance in colorectal cancer. *Redox Biol*. 2024;73:103207.
30. Lee J, Roh JL. SLC7A11 as a gateway of metabolic perturbation and ferroptosis vulnerability in cancer. *Antioxidants*. 2022. <https://doi.org/10.3390/antiox11122444>.
31. Lee J, You C, Kwon G, Noh J, Lee K, Kim K, et al. Integration of epigenomic and transcriptomic profiling uncovers EZH2 target genes linked to cysteine metabolism in hepatocellular carcinoma. *Cell Death Dis*. 2024;15(11):801.
32. Sen U, Coleman C, Gandhi N, Jethalia V, Demircioglu D, Elliott A, et al. SCD1 inhibition blocks the AKT-NRF2-SLC7A11 pathway to induce lipid metabolism remodeling and ferroptosis priming in lung adenocarcinoma. *Cancer Res*. 2025.
33. Deng J, Lin X, Qin J, Li Q, Zhang Y, Zhang Q, et al. SPTBN2 suppresses ferroptosis in NSCLC cells by facilitating SLC7A11 membrane trafficking and localization. *Redox Biol*. 2024;70:103039.
34. Dornier E, Rabas N, Mitchell L, Novo D, Dhayade S, Marco S, et al. Glutaminolysis drives membrane trafficking to promote invasiveness of breast cancer cells. *Nat Commun*. 2017;8(1):2255.
35. Wang X, Chen Y, Wang X, Tian H, Wang Y, Jin J, et al. Stem cell factor SOX2 confers ferroptosis resistance in lung cancer via upregulation of SLC7A11. *Cancer Res*. 2021;81(20):5217–29.
36. Luo L, Zhao Q, Cui X, Dong S, Wang Y, Jiang N, et al. Male-specific lethal 1 (MSL1) promotes Erastin-induced ferroptosis in colon cancer cells by regulating the KCTD12-SLC7A11 axis. *Cell Death Dis*. 2025;16(1):281.
37. Zhang H, Ma T, Wen X, Jiang J, Chen J, Jiang J, et al. SIK1 promotes ferroptosis resistance in pancreatic cancer via HDAC5-STAT6-SLC7A11 axis. *Cancer Lett*. 2025;623:217726.
38. Chen Q, Zheng W, Guan J, Liu H, Dan Y, Zhu L, et al. SOCS2-enhanced ubiquitination of SLC7A11 promotes ferroptosis and radiosensitization in hepatocellular carcinoma. *Cell Death Differ*. 2023;30(1):137–51.
39. Miao M, Pan M, Chen X, Shen J, Zhang L, Feng X, et al. IL-13 facilitates ferroptotic death in asthmatic epithelial cells via SOCS1-mediated ubiquitinated degradation of SLC7A11. *Redox Biol*. 2024;71:103100.
40. Zhou Q, Yu H, Chen Y, Ren J, Lu Y, Sun Y. The CRL3(KCTD10) ubiquitin ligase-USP18 axis coordinately regulates cystine uptake and ferroptosis by modulating SLC7A11. *Proc Natl Acad Sci U S A*. 2024;121(28):e2320655121.
41. Tang J, Long G, Hu K, Xiao D, Liu S, Xiao L, et al. Targeting USP8 inhibits O-GlcNAcylation of SLC7A11 to promote ferroptosis of hepatocellular carcinoma via stabilization of OGT. *Adv Sci*. 2023;10(33):e2302953.
42. Gu Y, Albuquerque CP, Braas D, Zhang W, Villa GR, Bi J, et al. mTORC2 regulates amino acid metabolism in cancer by phosphorylation of the cystine-glutamate antiporter xCT. *Mol Cell*. 2017;67(1):128–38 e7.

43. Yang J, Zhou Y, Xie S, Wang J, Li Z, Chen L, et al. Metformin induces ferroptosis by inhibiting UFMylation of SLC7A11 in breast cancer. *J Exp Clin Cancer Res.* 2021;40(1):206.
44. Fenczik CA, Zent R, Dellos M, Calderwood DA, Satriano J, Kelly C, et al. Distinct domains of CD98hc regulate integrins and amino acid transport. *J Biol Chem.* 2001;276(12):8746–52.
45. Fort J, de la Ballina LR, Burghardt HE, Ferrer-Costa C, Turnay J, Ferrer-Orta C, et al. The structure of human 4F2hc ectodomain provides a model for homodimerization and electrostatic interaction with plasma membrane. *J Biol Chem.* 2007;282(43):31444–52.
46. Pfeiffer R, Spindler B, Loffing J, Skelly PJ, Shoemaker CB, Verrey F. Functional heterodimeric amino acid transporters lacking cysteine residues involved in disulfide bond. *FEBS Lett.* 1998;439(1–2):157–62.
47. Fifield AL, Hanavan PD, Faigel DO, Sergienko E, Bobkov A, Meurice N, et al. Molecular inhibitor of QSOX1 suppresses tumor growth in vivo. *Mol Cancer Ther.* 2020;19(1):112–22.

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